

Global Warming: An Overview

What is Global Warming?

Global warming refers to the long-term rise in Earth's average surface temperature, driven primarily by the accumulation of greenhouse gases in the atmosphere. Since the pre-industrial era around 1850, global average temperatures have increased by approximately 1.1 degrees Celsius. The Intergovernmental Panel on Climate Change (IPCC) has confirmed that this warming trend is unequivocally linked to human industrial activity, not natural climate variability.

Greenhouse Gases

The primary greenhouse gases responsible for warming are carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), and fluorinated gases. Carbon dioxide is the most significant, accounting for about 76% of total greenhouse gas emissions. Atmospheric CO₂ has risen from around 280 ppm before industrialisation to over 421 ppm as of 2023, the highest level in at least 3 million years. Methane, though shorter-lived, is roughly 28 times more potent than CO₂ over a 100-year period.

Main Causes

Burning fossil fuels for energy and transport is the largest source, contributing approximately 75% of global greenhouse gas emissions. Deforestation adds another 10 to 12%, as trees that once absorbed CO₂ are cleared, mainly for agriculture. The agriculture sector itself, through livestock farming and fertiliser use, contributes a further 10 to 12% of emissions. Industrial processes such as cement and steel production account for roughly 5% of global emissions.

Observed Impacts

The consequences of global warming are already visible across the planet. Global sea levels have risen by about 20 cm since 1900 and are currently rising at 3.6 mm per year. Arctic sea ice extent in September has declined by around 13% per decade. Extreme heat events are now five times more likely than in the pre-industrial era. Ocean acidification has increased by 30% as oceans absorb more CO₂. Glaciers worldwide are retreating, threatening freshwater supplies for hundreds of millions of people who depend on glacier-fed rivers.

Tipping Points

Scientists warn about climate tipping points, which are thresholds beyond which changes become self-sustaining and irreversible. Examples include the potential collapse of the West Antarctic Ice Sheet, which could raise sea levels by more than 3 metres, and the thawing of permafrost regions that hold approximately 1.5 trillion tonnes of frozen organic carbon. A 2022 study identified 16 major tipping points and found that five could already be triggered at the current level of warming, and that tipping points can cascade and amplify one another.

Mitigation Strategies

Mitigation refers to actions that reduce or prevent greenhouse gas emissions. The IPCC concludes that limiting warming to 1.5 degrees Celsius requires reaching net-zero CO₂ emissions globally by around 2050. The transition to renewable energy is central to this goal. Solar and wind power are now the cheapest sources of new electricity generation in most regions. Global renewable electricity capacity grew by 295 GW in 2022 alone. Improving energy efficiency in buildings, industry, and transport is also critical, with the International Energy Agency estimating that efficiency measures could reduce global energy demand by 40% by 2050.

Technological carbon removal approaches such as Direct Air Capture are being developed and scaled. As of 2023, these technologies capture around 10,000 tonnes of CO₂ per year globally, far below the 6 to 10 billion tonnes per year that most 1.5 degree scenarios require by 2050. Preserving and restoring forests and other natural ecosystems also plays a vital role, as natural carbon sinks currently absorb around 55% of annual human emissions.

International Policy

The Paris Agreement, signed in 2015, established a global framework to limit warming to well below 2 degrees Celsius, with efforts toward 1.5 degrees. Over 140 countries have committed to net-zero targets. However, current national pledges fall short of what is needed, placing the world on a trajectory of approximately 2.5 to 2.9 degrees of warming by 2100. Carbon pricing mechanisms now cover around 23% of global greenhouse gas emissions.

Adaptation

Adaptation involves adjusting to the effects of climate change that are already unavoidable. Examples include building sea walls and coastal defences, developing drought-resistant crop varieties, creating early warning systems for extreme weather, and planting urban forests to cool cities. The IPCC estimates that adaptation costs for developing nations alone could reach 280 to 500 billion dollars per year by 2050.

Climate Justice and Economic Impacts

Global warming does not affect everyone equally. Low-lying island nations, Sub-Saharan Africa, South Asia, and Arctic communities face the most severe impacts despite contributing the least to global emissions. Economic modelling suggests that unchecked warming could reduce global GDP by 10 to 23% by 2100, with tropical regions suffering the greatest losses. Conversely, rapid decarbonisation is projected to generate 26 trillion dollars in economic benefits by 2030 through improved air quality, energy security, and technological innovation.

Conclusion

Global warming is the defining environmental challenge of the 21st century. The scientific evidence is clear and the impacts are already being felt worldwide. Addressing it requires an urgent transformation of energy systems, land use, and consumption patterns, backed by international cooperation and equitable climate finance. Every fraction of a degree of warming avoided reduces risk, and the decisions made in the coming decade will shape the climate for centuries to come.